AN F_P-ALGEBRA IS ABSOLUTELY WEAKLY NORMAL IF AND ONLY IF IT IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0EVV (sources/stacks-flat/pinned-original-source.tex line 12125).

Allowed standard prerequisites (editorial).

- Seminormal and absolutely weakly normal rings and condition (2)(b) of the definition. Morphisms, Definition labelled morphisms-definition-seminormal-ring (Stacks tag 0EUL), invoked at proof lines 12131-12133. Supplies the definition of absolute weak normality, in particular condition (2)(b) used for the immediate direction, and the two root existence conditions verified in the counted span.
- Perfect F_p-algebras and the Frobenius being an isomorphism. Algebra, the notion of a perfect ring of characteristic p and the resulting existence of p-th roots. Supplies that every x in A has a p-th root a with a^p = x, which is what makes the case analysis at proof lines 12137-12152 work.

Source reference. Original definition 0EUL: sources/stacks-chow/tags.txt, source lines 18556–18556. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12132 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let p be a prime number. An $\mathbf{F}_p$-algebra A is absolutely weakly normal if and only if it is perfect.*

Proof. It is immediate from condition (2)(b) in Morphisms, Definition 0EUL that if A is absolutely weakly normal, then it is perfect.

Assume A is perfect. Suppose $x, y \in A$ with $x^3 = y^2$. If $p > 3$ then we can write $p = 2n + 3m$ for some $n, m > 0$. Choose $a, b \in A$ with $a^p = x$ and $b^p = y$. Setting

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$c = a^n b^m$ we have

$$c^{2p} = x^{2n} y^{2m} = x^{2n+3m} = x^p$$

and hence $c^2 = x$. Similarly $c^3 = y$. If $p = 2$, then write $x = a^2$ to get $a^6 = y^2$ which implies $a^3 = y$. If $p = 3$, then write $y = a^3$ to get $x^3 = a^6$ which implies $x = a^2$.

Suppose $x, y \in A$ with $\ell^\ell x = y^\ell$ for some prime number ℓ. If $\ell \neq p$, then $a = y/\ell$ satisfies $a^\ell = x$ and $\ell a = y$. If $\ell = p$, then $y = 0$ and $x = a^p$ for some a. $\square$